

55. Jump game:

You are given an integer array `nums`. You are initially positioned at the array's **first index**, and each element in the array represents your maximum jump length at that position.

Return `true` if you can reach the last index, or `false` otherwise.

Example 1:

Input: `nums = [2,3,1,1,4]`

Output: `true`

Explanation: Jump 1 step from index 0 to 1, then 3 steps to the last index.

Example 2:

Input: `nums = [3,2,1,0,4]`

Output: `false`

Explanation: You will always arrive at index 3 no matter what. Its maximum jump length is 0, which makes it impossible to reach the last index.

Constraints:

- `1 <= nums.length <= 104`
- `0 <= nums[i] <= 105`

Solution: Time = $O(n)$ and Space = $O(1)$

```
class Solution:
    def canJump(self, nums: List[int]) -> bool:
        mxlvl = 0
        for i in range(len(nums)):
            if (i > mxlvl):
                return False
```

```
mxlvl = max(mxlvl, i + nums[i])  
  
return True
```